DAY 12 | POST-READ

React and APIs: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
&&	Render something or nothing. Beware of 0: use count > 0 &&.
Ternary	cond ? A : B to choose between two views.
Early return	if (loading) return ...; keeps the main view simple.
Lists	items.map(i =>). Keys must be stable and unique.
Derived data	Filter and sort while rendering; do not store computed copies in state.
useEffect	Runs after render. [] means once on mount. Use it to start fetching.
Fetch states	loading, error, empty and data. Design a screen for each.
POST from React	fetch with method, JSON headers and body; update state with the reply.
CORS	Needed when front end and API have different origins (ports count).

Part B: Going beyond today

1. Cleaning up effects

```
useEffect(() => {
  const controller = new AbortController();
  fetch(url, { signal: controller.signal }).then(...);
  return () => controller.abort();   // runs if the component disappears
}, [url]);
```

2. Effects that depend on values

useEffect(fn, [query]) re-runs whenever query changes, for example to search as the user types. Combine with a short delay (debounce) so you do not call the API on every keystroke.

3. A tiny custom hook

```
function useFetch(url) {
  const [state, setState] = useState({ loading: true, data: null, error: null });
  useEffect(() => { ... }, [url]);
  return state;
}
```

Custom hooks package reusable logic, much like FastAPI dependencies.

4. Optimistic updates

Update the screen immediately when the user toggles a task, then call the API. If the call fails, undo the change and show an error. Apps feel much faster this way.

5. Data fetching libraries

In larger apps, libraries such as TanStack Query handle caching, retries and refreshing for you. Learn the plain useEffect way first, as you did today.

Part C: Check yourself

- Why can {count && Items} show a stray 0?
- What problem do keys solve?
- When does useEffect(fn, []) run?
- Name the four states every data screen should handle.
- Why does a React app on port 5173 need CORS to call FastAPI on port 8000?

Part D: Practice ideas for tonight

- Add an edit mode to the Task Board: double click a title to edit it, then PATCH it.
- Add a Retry button to the error state.
- Build a page that lists products from your Day 9 products API with pagination buttons.

Coming up on Day 13: testing with PyTest and React tests, then an introduction to data engineering: the data lifecycle, ETL vs ELT and ingestion.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today